

ICCAD 2026 #707 — Supplementary Results for Author Response

Measured Layer Sensitivity Guides Mixed-Precision ADC Allocation for CIM-Based LLM Inference

These results support the rebuttal only; the paper is unchanged. Unless noted, the setup is OPT-125M, WikiText-2, p99-clip ADC calibration, paper data split, INT8 per-channel weights. “ Δ PPL/layer” is the per-layer measured sensitivity when a group’s ADC bit-width is reduced. We report *denoised, fixed-seed* measurements; *absolute* Δ PPL magnitudes depend on the calibration protocol and the number of profiling batches, but the allocation-relevant quantities — that W_{fc2} is the most ADC-sensitive group and that the FFN aggregate exceeds the attention aggregate — are reproduced. We report group/aggregate quantities; the single-layer W_{head} group is excluded as numerically unstable (the paper likewise reports the most-sensitive *non- W_{head}* layer, Sec. 5.4).

Table S1. Metric invariance (PPL vs NLL) and weight-quant robustness (OPT-125M, INT8 per-channel weights, probe 7→6). The FFN aggregate exceeds the attention aggregate, and W_{fc2} is the most-sensitive group, under *both* PPL- and NLL-based sensitivity. Because $PPL = \exp(NLL)$ is strictly monotonic with a shared baseline, the full per-group ranking is provably metric-independent: empirically the PPL and NLL rankings are identical (Spearman $\rho = 1.0$) and the group-level ILP allocation is unchanged. (Answers #707C-Q1; supports CR-2 / #707C-C7 / #707D-Q1: the ordering survives a stronger weight-quant floor.)

Sensitivity metric	FFN aggregate	Attention aggregate
Δ PPL/layer (PPL-based)	+0.070	+0.030
Δ NLL/layer ($\times 10^{-4}$, NLL-based)	+2.13	+0.93
W_{fc2} is the single most-sensitive group under both metrics.		
PPL vs NLL ranking: Spearman $\rho = 1.00$; group-level ILP allocation identical.		
Values use the fixed paper-split seed; Table S2 row 1 reports the 3-seed mean.		

Table S2. Robustness of the FFN > attention ordering (aggregate FFN vs aggregate attention mean Δ PPL/layer; INT8 per-channel weights). Swept over probe depth and calibration seed on OPT-125M. FFN > attention holds in 7/9 single-seed OPT settings; seed-averaged it holds clearly at 7→6 (+0.044 vs +0.023) and overwhelmingly at 7→4 ($\approx 11\times$); with a larger profiling budget (calib 16, eval 60) the 7→6 ordering is stable across seeds, indicating the small-probe single-seed noise is a finite-batch artifact. (Supports CR-1/CR-3; answers #707A-W1, #707C-Q2, #707D-W3.)

Model	Probe	FFN Δ PPL/l	Attn Δ PPL/l	FFN > Attn
OPT-125M	7→6 (seed mean)	+0.044	+0.023	yes
OPT-125M	7→6 (3 seeds)	2/3 single-seed; 2/2 at calib16/eval60		2/3
OPT-125M	7→5 (3 seeds)	noisy		2/3
OPT-125M	7→4 (3 seeds)	+0.67 (mean)	+0.063 (mean)	3/3
OPT overall 7/9; clear at 7→6 seed-mean and 7→4.				

Table S3. Full profiling→ILP→PPA pipeline on Qwen2-7B at a usable operating point (WikiText-2; *max-clip* ADC calibration — one of the paper’s three calibration schemes (Table I) — 8-bit nominal; baseline PPL ≈ 16 vs clean ≈ 10). Profiling-guided ILP allocates ADC bits from the measured group sensitivities under an ADC-area budget; ADC energy follows the same 2^b model as in the paper. FFN is far more ADC-sensitive than attention, so the ILP protects the sensitive group and reduces the insensitive high-column-count group. Downstream accuracy (PIQA/BoolQ) is preserved within ≈ 1.6 pp of the CIM-8b baseline. (Answers #707C-C8 and #707A-W2: a complete modern-architecture pipeline at a realistic operating point.)

Configuration	ADC area+energy		PPL	PIQA	BoolQ
Clean (fp16, no CIM)	—		≈ 10	80.4	85.2
CIM uniform 8b (baseline)	—		15.6	77.4	75.0
Profiling-ILP (34% budget)	34%	15.9	(+1.8%)	77.2	73.4
Profiling-ILP (52% budget)	52%	16.4	(+4.9%)	—	—
Proxy-blind (reduce the <i>most</i> -sensitive group): PPL $\rightarrow$ 1938 (model destroyed).					